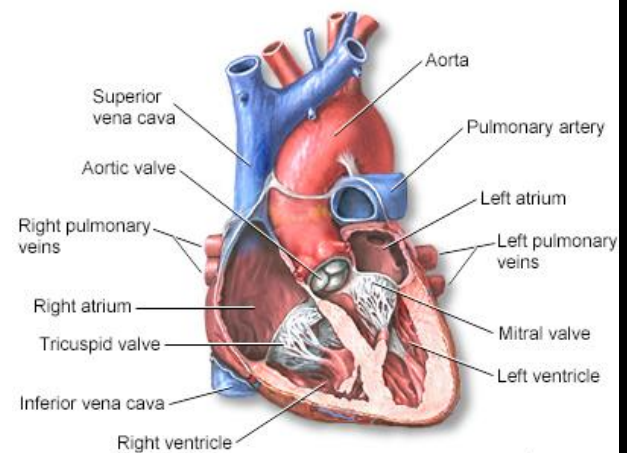


1.INTRODUCTION:-

The Heart is one of the most important organs in the human body. It is the centre of the circulatory system. The heart functions as a pump that propels blood to different parts of the human body through a network of blood vessels, supplying a constant supply of oxygen as well as other vital nutritional components. If the heart ever stops functioning and ceases to pump blood, the body will shut down and within very less time a person will expire.

1.1 What is coronary heart disease?

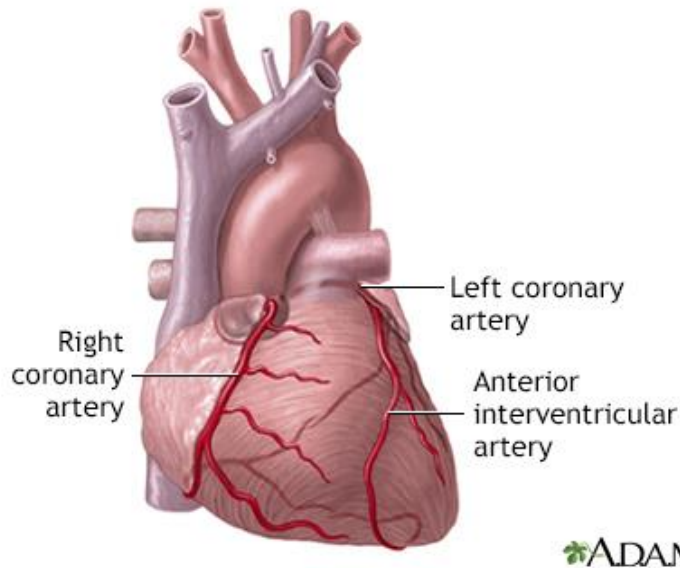
Coronary heart disease (CHD) is also called as coronary artery disease. The coronary arteries are the blood vessels that carry blood to your heart. Coronary artery disease is the narrowing or blockage of the coronary arteries. This condition is usually caused by atherosclerosis. Atherosclerosis is the build-up of cholesterol and fatty deposits (called plaques) inside the arteries. These plaques can clog the arteries or damage the arteries, which limits or stops blood flow to the heart muscle.



ADAM

If the heart does not get enough blood, it cannot get the oxygen and nutrients it needs to work properly. This can cause chest pain (angina) or a heart attack.

Coronary artery disease (CAD) is the most common type of heart disease in the United States. It is sometimes called ischemic heart disease.



Coronary arteries supply blood to the heart muscle. The right coronary artery supplies both the left and right heart, the left coronary artery supplies the left heart.

1.1.1 Causes

Coronary artery disease is thought to begin with damage or injury to the inner layer of a coronary artery, sometimes as early as childhood. The damage may be caused by various factors, including:

- Smoking
- High blood pressure
- High cholesterol
- Diabetes or insulin resistance

- Not being active (sedentary lifestyle)

Once the inner wall of an artery is damaged, fatty deposits (plaque) made of cholesterol and other cellular waste products tend to collect at the site of injury. This process is called atherosclerosis. If the plaque surface breaks or ruptures, blood cells called platelets clump together at the site to try to repair the artery. This clump can block the artery, leading to a heart attack.

1.1.2 Symptoms

An acute coronary event, such as a heart attack, may cause the following symptoms:

- Cold sweats
- Dizziness
- Light-headedness
- Nausea or a feeling of indigestion
- Neck pain
- Shortness of breath, especially with activity
- Sleep disturbances
- Weakness

The usage of information technology in health care industry is increasing day by day to aid doctors in decision making activities. It helps doctors and physicians in disease management, medications and discovery of patterns and relationships among diagnosis data. Current approaches to predict cardiovascular risk fail to identify many people who would benefit from preventive treatment, while others receive unnecessary intervention. Machine-learning offers opportunity to

improve accuracy by exploiting complex interactions between risk factors. We assessed whether machine-learning can improve cardiovascular risk prediction.

1.2 What is Deep learning?

Deep Learning (DL) is a subfield of **Machine Learning (ML)** that uses **Artificial Neural Networks (ANNs)** with multiple layers (hence "deep") to model complex patterns in data. Unlike traditional ML algorithms, which require manual feature extraction, deep learning models **automatically learn hierarchical representations** from raw data, making them highly effective for tasks like image recognition, natural language processing (NLP), and medical diagnosis.

Deep learning has gained prominence due to:

- Increased computational power (GPUs, TPUs)
- Availability of large datasets (Big Data)
- Advancements in neural network architectures

1.2.1 Application of Deep learning

- Deep Learning (DL) has revolutionized various industries by enabling machines to learn complex patterns from vast amounts of data. Below are some key applications of deep learning, with a focus on healthcare, biomedical research, and other domains.

1. Healthcare & Medical Applications

(1) Disease Diagnosis & Prediction

- Medical Imaging Analysis
 - Radiology (X-ray, CT, MRI): Detecting tumors, fractures, and abnormalities using CNNs.

- Ophthalmology: Diagnosing diabetic retinopathy and glaucoma from retinal scans.
- Pathology: Identifying cancer cells in histopathology slides.
- Cardiovascular Disease Prediction
 - Predicting coronary heart disease (CHD) using patient history (ANN, CNN-1D).
 - Analyzing ECG signals with 1D-CNNs/LSTMs for arrhythmia detection.

(2) Drug Discovery & Genomics

- Molecular Property Prediction: Using Graph Neural Networks (GNNs) to predict drug interactions.
- Protein Folding: AlphaFold (DeepMind) predicts 3D protein structures.
- Personalized Medicine: Tailoring treatments based on genetic data (Deep Learning + Bioinformatics).

(3) Wearable & Remote Patient Monitoring

- Real-time Health Tracking:
 - Detecting atrial fibrillation (AFib) from smartwatch ECG data.
 - Predicting seizures in epilepsy patients using LSTM networks.

2. Computer Vision & Image Processing

- Autonomous Vehicles:
 - Object detection (YOLO, Faster R-CNN) for self-driving cars.
- Facial Recognition & Biometrics:
 - Used in security systems (e.g., Face ID).
- Satellite & Aerial Imaging:

- Deforestation tracking, disaster response, and urban planning.

3. Natural Language Processing (NLP)

- Chatbots& Virtual Assistants:
 - GPT-4, ChatGPT, and medical chatbots for patient queries.
- Clinical Text Analysis:
 - Extracting insights from Electronic Health Records (EHRs) using BERT/Transformers.
- Automated Medical Transcription:
 - Converting doctor-patient conversations into structured notes.

4. Finance & Business

- Fraud Detection:
 - Detecting credit card fraud using Autoencoders (anomaly detection).
- Algorithmic Trading:
- Predicting stock trends with LSTMs & Reinforcement Learning (RL).
- Customer Sentiment Analysis:
- Analyzing social media/reviews using NLP models.
-

5. Industrial & IoT Applications

- Predictive Maintenance:
 - Detecting machinery failures using CNNs on sensor data.
- Smart Agriculture:
 - Crop disease detection from drone images.

-
- Customer Sentiment Analysis:
 - Analyzing social media/reviews using NLP models.
- 6. Emerging Applications**
 - Metaverse& Augmented Reality (AR):
 - DL-powered virtual avatars and real-time object recognition.
 - Climate Science:
 - Weather prediction using Physics-Informed Neural Networks (PINNs).

1.2.2 Types of Deep Learning

(1) Artificial Neural Networks (ANNs)

- **Basic structure:** Input layer, hidden layers, and output layer.
- **Used for:** Regression, classification, and function approximation.
- **Example:** Predicting coronary heart disease (CHD) risk from structured clinical data.

(2) Convolutional Neural Networks (CNNs)

- **Specialized for:** Grid-like data (images, 1D signals, time-series).
- **Key layers:** Convolutional layers (feature extraction), pooling layers (dimensionality reduction), and fully connected layers (classification).
 - **2D-CNN:** Used in image processing (e.g., X-ray, MRI analysis).
 - **1D-CNN:** Processes sequential data (e.g., ECG signals, clinical time-series).
- **Example:** Detecting CHD from ECG signals using 1D-CNN.

(3) Recurrent Neural Networks (RNNs) & Long Short-Term Memory (LSTM)

- **Designed for:** Sequential/temporal data (time-series, text, speech).

- **LSTM:** A variant of RNN that solves the vanishing gradient problem, useful for long-term dependencies.
- **Example:** Predicting heart disease progression from patient history.

(4) U-Net (Originally for Image Segmentation)

- **Architecture:** Encoder-decoder with skip connections (preserves spatial details).
- **Modified for 1D:** Can be adapted for sequential medical data (e.g., predicting CHD risk).
- **Example:** Segmenting coronary arteries in angiograms or analyzing risk factor sequences.

(5) Autoencoders (AEs) & Variational Autoencoders (VAEs)

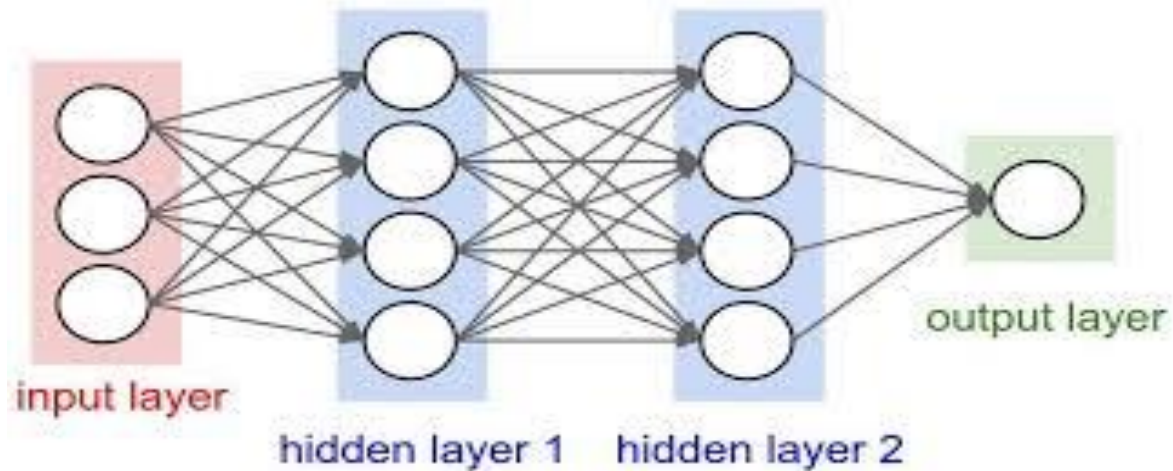
- **Purpose:** Unsupervised learning, dimensionality reduction, anomaly detection.
- **Used in:** Feature extraction from high-dimensional medical data.

(6) Transformer Models (e.g., BERT, Vision Transformer)

- **Revolutionized NLP and image processing** using self-attention mechanisms.
- **Example:** Analyzing electronic health records (EHR) for CHD prediction.

1.3 ANN (Artificial Neural Network)

ANNs are computational models inspired by the human brain, consisting of interconnected layers of nodes (neurons). They form the foundation of deep learning by enabling machines to learn complex patterns and representations.



Structure

- **Input Layer:** Receives raw data (features).
- **Hidden Layers:** Perform computations and transformations. Deep networks have multiple hidden layers.
- **Output Layer:** Produces final predictions or classifications.

Neuron Function

Each neuron computes:

$$z = w_1 * x_1 + w_2 * x_2 + \dots + w_n * x_n + b$$

$$a = \text{activation}(z)$$

Where:

- w = weights
- x = input
- b = bias
- a = output (activation)

1.3.1 Activation Functions:

Used to introduce non-linearity:

- **Sigmoid**
- **Tanh:**
- **ReLU** (Rectified Linear Unit)
- **Leaky ReLU, Softmax** (for classification)

Sigmoid activation function:

The **Sigmoid function** maps any real-valued number into a value between 0 and 1. It's commonly used in binary classification tasks.

Mathematical Formula:

$$\sigma(x) = 1/(1+e^{-x})$$

Where:

- x is the input to the neuron,
- e is the base of the natural logarithm.

Advantages

- **Smooth and differentiable**, which is essential for gradient-based optimization.
- **Outputs can be interpreted as probabilities**, especially in the output layer for binary classification.
- Historically used in early neural networks.

Disadvantages:-

- **Vanishing Gradient Problem:** For large positive or negative inputs, the gradient becomes very small, which slows down training.
- **Output not zero-centered:** Can cause issues in convergence due to zig-zagging updates in gradient descent.
- **Not ideal for deep networks:** ReLU and its variants are preferred in hidden layers.

Tanh Activation Function:-

The **Tanh** function maps input values to an output in the range **(-1, 1)**.

It is a scaled and shifted version of the sigmoid function.

$$\tanh(x) = (e^x + e^{-x}) / (e^x - e^{-x})$$

- **S-shaped curve**, similar to sigmoid but centered at zero.
- **Range:** (-1, 1)
- **Zero-centered:** Output is symmetric about the origin, which helps in training compared to sigmoid.

Advantages:-

- **Zero-centered outputs:** Helps gradients flow more evenly, leading to faster convergence than sigmoid.
- Suitable for hidden layers in shallow networks.
- Smooth and differentiable.

Disadvantages:-

- **Vanishing Gradient Problem:** Like sigmoid, gradients become very small for large input values, making learning slow in deep networks.

- ReLU and its variants are often better in deeper models.

Often used in **hidden layers** of neural networks, especially before ReLU became popular.

Sometimes still used in **RNNs** (e.g., for gating mechanisms).

RELU ACTIVATION FUNCTION:

The ReLU function is defined as:

$$\text{ReLU}(x) = \max(0, x) \quad \text{ReLU}(x) = \max(0, x)$$

This means:

- If $x > 0$, then $\text{ReLU}(x) = x$
- If $x \leq 0$, then $\text{ReLU}(x) = 0$

Graph:

The graph of ReLU looks like a line that is flat at 0 for all negative inputs and increases linearly for positive inputs.

Advantages:

- **Computationally efficient:** Simple to compute.
- **Reduces likelihood of vanishing gradients:** Compared to sigmoid or tanh, which can saturate.
- **Sparse activation:** Activates only a subset of neurons, which makes the network more efficient.

Disadvantages:

- **Dying ReLU problem:** Neurons can "die" during training if they only output zero (i.e., never activate), especially with high learning rates.

Softmax activation function:

The **Softmax activation function** is commonly used in the output layer of neural networks for **multi-class classification problems**. It converts a vector of raw scores (called logits) into a vector of probabilities, where each probability represents the likelihood that a particular input belongs to a specific class.

- **Output range:** (0, 1)
- **Sum of outputs:** 1 (i.e., outputs form a probability distribution)
- **Use case:** Typically used in the final layer of a classifier that needs to assign probabilities across **multiple classes**.

Training an ANN:

- **Forward Propagation:** Compute predictions.
- **Loss Function:** Measures error (e.g., MSE, Cross-Entropy).
- **Backpropagation:** Computes gradients of the loss w.r.t weights.
- **Gradient Descent** (or variants like Adam): Optimizes weights using gradients.

1.4 Convolutional Neural Networks (CNNs)

What is a CNN?

A **Convolutional Neural Network (CNN)** is a type of deep learning model designed to process **grid-like data**, especially images. CNNs automatically detect and learn **spatial hierarchies** in visual data.

Convolutional Layer

- Applies **filters/kernels** to the input image to extract features like edges, textures, patterns.
- Produces a **feature map**.
- Each filter learns a different feature.

Operation:

Feature Map = Input * Filter + Bias

Activation Function (ReLU)

- Applied after convolution to introduce **non-linearity**.
- ReLU: $f(x) = \max(0, x)$

Pooling Layer

- Reduces spatial dimensions (downsampling) to:
 - Decrease computation
 - Reduce overfitting
 - Retain important features

Types:

- Max Pooling
- Average Pooling

Fully Connected Layer

- Final layers where all neurons are connected to every output from the previous layer.
- Combines features and performs classification.

Applications

- Image Classification (e.g., Cats vs Dogs)
- Object Detection (e.g., YOLO, SSD)
- Face Recognition
- Medical Imaging
- Self-driving Cars

1.5 1D Convolutional Neural Networks (1D CNN)

1D Convolution Layer

- Slides a 1D filter over the sequence to detect local patterns.
- Common patterns: spikes, trends, repetitions.
- Parameters: number of filters, kernel size, stride, padding.

Pooling Layer (1D)

Down samples the sequence length to reduce dimensionality and retain dominant features.

Flatten & Dense Layers

Flatten & Dense Layers :

After several Conv1D + Pooling layers, data is flattened and passed to fully connected layers for classification or regression.

1.6 U-Net Algorithm :

U-Net is a Convolutional Neural Network (CNN) architecture developed for semantic segmentation — where the goal is to classify each pixel in an image into a category (e.g., background vs object).

- Originally proposed in 2015 for biomedical image segmentation.

- Named "U-Net" because of its U-shaped architecture.

U-Net Architecture Overview

The U-Net consists of two main parts:

A. Contracting Path ("Encoder")

- Similar to a typical CNN.
- Repeated application of:
 - Conv2D + ReLU
 - MaxPooling
- Purpose: Extract context (what is in the image).

Example:

- Conv $\rightarrow$ ReLU $\rightarrow$ Conv $\rightarrow$ ReLU $\rightarrow$ MaxPooling $\rightarrow$ Repeat

B. Expanding Path ("Decoder")

- Upsamples the features using:
 - Up-convolution (Transpose Conv) or Upsampling
 - Concatenation with corresponding feature map from encoder (skip connection)
- Purpose: Recover spatial information for precise localization.

2 Objective

Heart problems have gained a lot of interest in medical research because of their impact on human health where early diagnosis is critical to delaying the development of heart disease, the world's leading cause of death. thus, it is much needed to predict the possibility of occurrence of heart disease based on their attributes.

This research aims into a variety of deep learning algorithms for predicting heart disease using Framingham dataset.

3. Literature Survey

Current approaches To predict heart disease

1. Heart Disease Prediction Using Machine Learning and Data Mining Techniques: Application of Framingham Dataset. Turkish Journal of Computer and Mathematics Education. Vol.12 No.14(2021), 4864-4870. Walaa Adel Mahmoud , Prof. Dr. Mohamed Aborizka , Prof. Dr. Fathy Ahmed Elsayed Amer²

In this research work The 10-fold cross-validation resampling is used to validate the prediction model. 5 different machine learning algorithms (KNearestNeighbors, LogisticRegression, decisionTree, Random Forest and support vector machine) are used in this work. Accuracy scores of each algorithm are evaluated.

KNearestNeighbors-83.95 %

SupportVectorMachine-84.5 %

DecisionTree-84.82 %

LogisticRegression-84.89 %

RandomForest-85.05 %

The Random Forest algorithm is found to have highest accuracy.

Remark

In this approach the features are not standardized.

2. Machine Learning- Based Classification Algorithms for the Prediction of Coronary Heart Diseases.

Kelvin Kwakye & Emmanuel Dadzie

Remark

In this approach missing values are handled by mean values

4.Problem definition

4.1 Task definition

In this project the task is to predict whether a person is going to have coronary heart disease or not in upcoming 10 years by using 3 popular deep learning algorithms. These are ANN algorithm, 1D CNN and UNET algorithm. After implementing these algorithms, the best model will be found for heart disease prediction. A graphical user interface will be designed for to make prediction for new dataset.

4.2 Heart disease data set

This dataset is located at Kaggle website. The dataset contains 16 attributes in all, of which 15 are independent factors and one variable i.e., Ten Year CHD is the dependent variable or target variable. This data set contains total 4240 data. All attributes in data set are numeric.

The 16 attributes that are defined here are as follows

Demographic:

- Sex: male or female (1-male and 0-female)
- Age: Age of the patient;(Continuous - Although the recorded ages have been truncated to whole numbers, the concept of age is continuous)
- Education: no further information provided

Behavioural:

- Current Smoker: whether or not the patient is a current smoker (Nominal)
- Cigs Per Day: the number of cigarettes that the person smoked on average in one day (can be considered continuous as one can have any number of cigarettes, even half a cigarette.)

Information on medical history:

- BP Meds: whether or not the patient was on blood pressure medication (Nominal)
- Prevalent Stroke: whether or not the patient had previously had a stroke (Nominal)
- Prevalent Hyp: whether or not the patient was hypertensive (Nominal)
- Diabetes: whether or not the patient had diabetes (Nominal)

Information on current medical condition:

- Tot Chol: total cholesterol level (Continuous)
- Sys BP: systolic blood pressure (Continuous)
- Dia BP: diastolic blood pressure (Continuous)
- BMI: Body Mass Index (Continuous)
- Heart Rate: heart rate (Continuous - In medical research, variables such as heart rate though in fact discrete, yet are considered continuous because of large number of possible values.)
- Glucose: glucose level (Continuous)

Target variable to predict:

- 10year risk of coronary heart disease (CHD) - (binary: “1”, means “Yes”, “0” means “No”)

Attribute & description	
Age	(32-70)
Sex	0=Female, 1=Male
Education	It takes values as: 1=High School, 2=High School or GED, 3=College or Vocational School, 4=College
Current Smoker	0=No 1=Yes
Cigs Per Day	Number of Cigarettes smoked Per Day (0-70)
BP Meds	0=No 1=Yes
Prevalent Stroke	0=No 1=Yes
Prevalent Hyp	0=No 1=Yes
Diabetes	0=No 1=Yes
Tot Chol	Serum Cholesterol (107-696) (mg/dl)
Sys BP	(83.5-295) (mm/hg)
Día BP	(48-142.5) (mm/hg)
Body Mass Index	(15.54-56.8)
Heart Rate	Heart Rate achieved (44-143)
Glucose	(40-394) (mg/dl)
10-year CHD	0=Healthy, 1=Diseases

5. Experimental set up and evaluation

Python:

Python is an interpreted, high-level, general-purpose programming language. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace. It provides constructs that enable clear programming on both small and large scales. In July 2018, Van Rossum stepped down as the leader in the language community.

Python features a dynamic type system and automatic management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.^[29]

Python interpreters are available for many operating systems. C Python, the reference implementation of Python, is open source software and has a community-based development model, as do nearly all of Python's other implementations. Python and C Python are managed by the non-profit Python Software Foundation.

Here, we have used Python 3.8 as it is the stable version as on now.

Python Packages:

They are simply directories, but with a twist. Each package in Python is a directory which **MUST** contain a special file called `__init__.py`. This file can be empty, and it indicates that the directory it contains is a Python package, so it can be imported the same way a module can be imported.

Packages are a way of structuring Python's module namespace by using "dotted module names". A.B stands for a sub module named B in a package named A. Two different packages like P1 and P2 can both have modules with the same name, let's say A, for example. The sub module A of the package P1 and the sub module A of the package P2 can be totally different. A package is imported like a "normal" module.

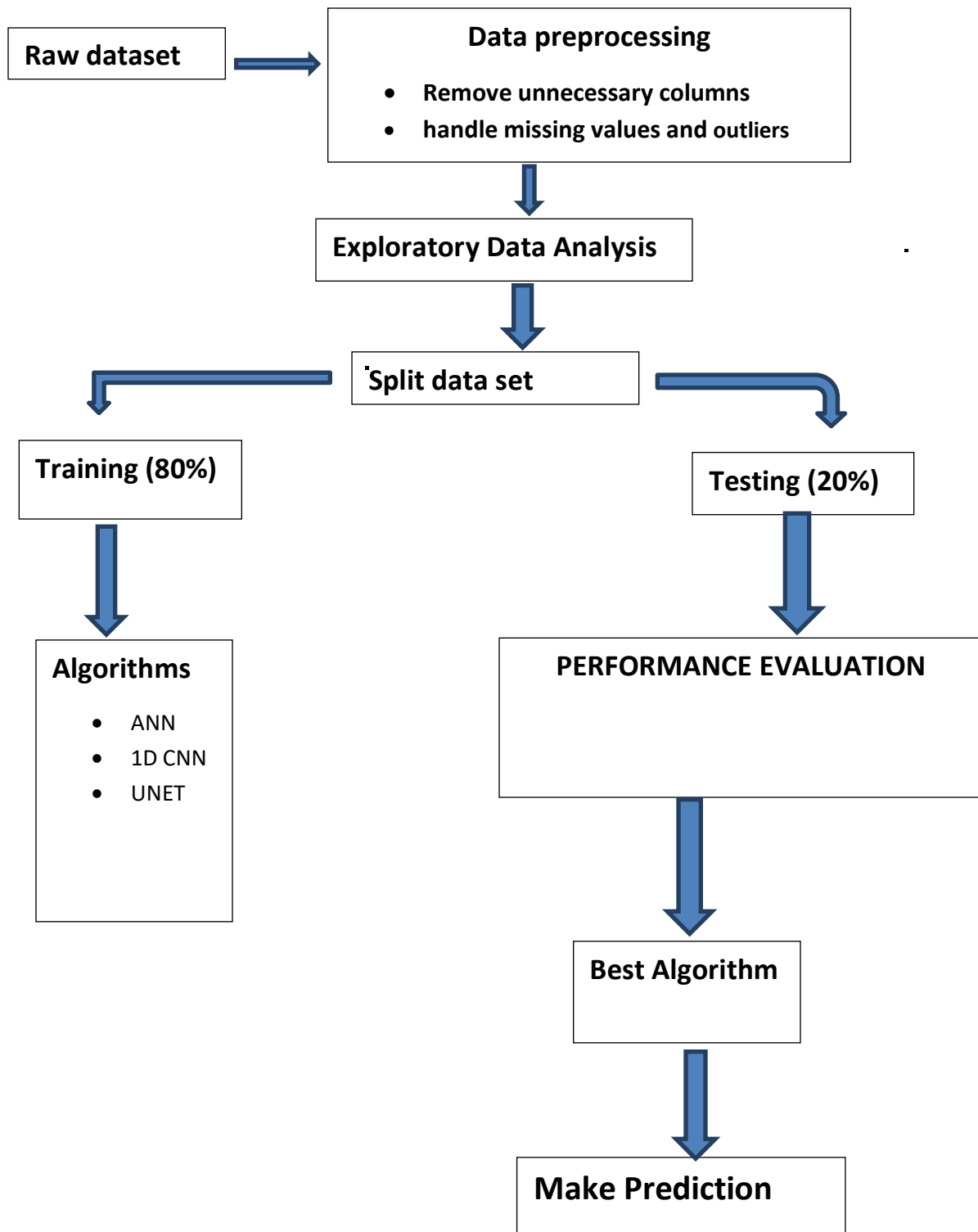
5.1 Components to be installed:

Anaconda (python 3.9)

Python Packages to be installed

- **pandas**- used to provide fast, flexible and expressive data structures to make working with “relational” or “labeled” data both easy and intuitive.
- **numpy**: To work with arrays
- **tkinter**- used for developing Graphical User Interface (GUI).
- **sklearn**- efficient for data mining and data analysis. It contains a number of algorithms which can be implemented directly when required.
- **matplotlib**- contains modules which are used for plotting graphs.
- **Seaborn**– used for Graphical Presentation
- **Tensorflow**—used for deep learning
- **Keras**– used for deep learning

5.2 Methodology



5.2.1 Data Preprocessing

Data preprocessing is a group of techniques that are applied on the data to improve the quality of the data. Raw data can be converted to useful data. It uses various techniques such as handling missing values, detecting outliers, removing unwanted attributes etc.

Once data is cleaned then it can be used in machine learning models for training. Data preprocessing is one of the most important steps involved in predictive modelling.

Steps followed in data preprocessing

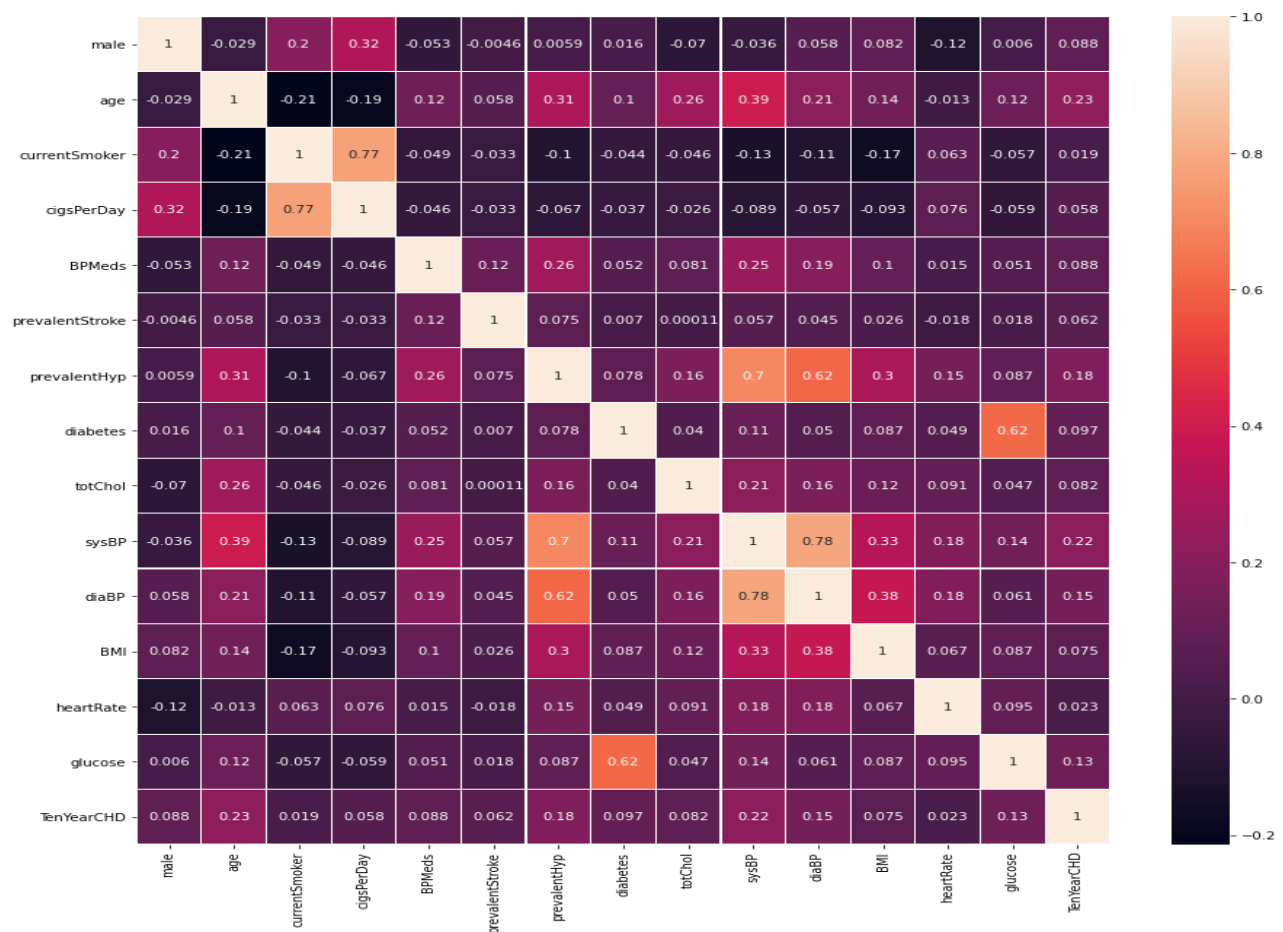
- Education column is removed as it has no relation with heart disease
- Most correlated features are found by using correlation matrix(heatmap).
Sysbp andDiabp are highly correlated
Current smoker and Cigsperday are highly correlated.so only one of them is selected.
- Missing values are handled by imputation .
- Outliers found and removed.
- Duplicated rows removed.
- Features have been standardized by using MInMaxScaler.

5.2.2 Exploratory analysis

Exploratory Data Analysis (EDA) is an approach to analyse the data using visual techniques. It is used to discover trends, patterns, or to check assumptions with the help of statistical summary and graphical representations.

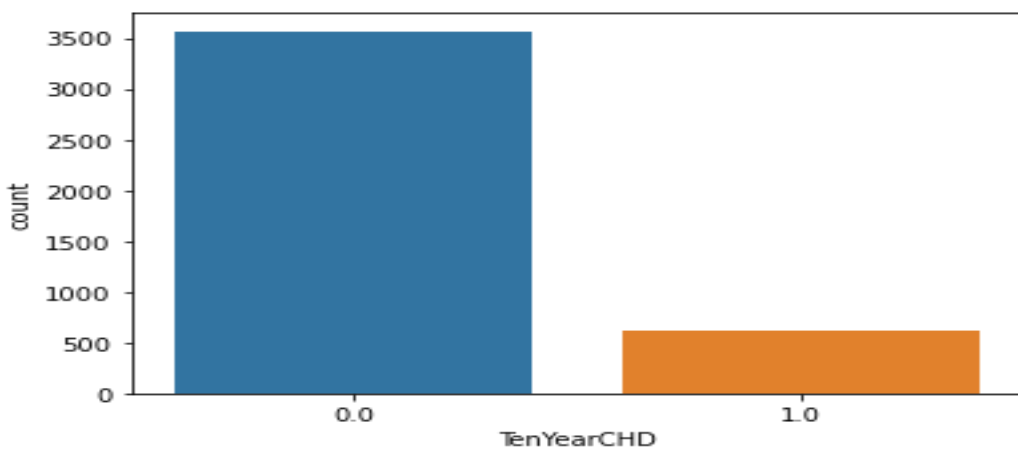
CorelationMatrix :

Corelation Matrix is used to show the relationship among the features. It is shown by heat map

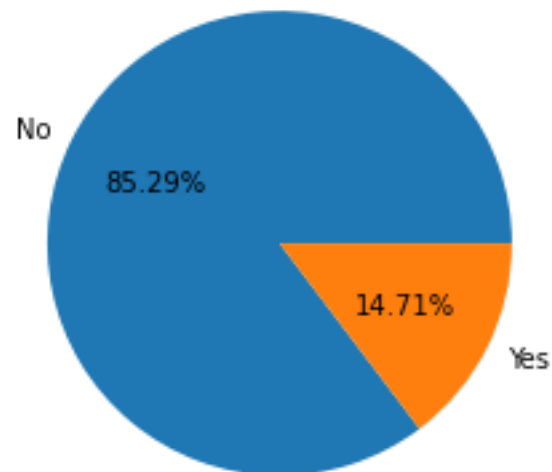


(Heat Map)

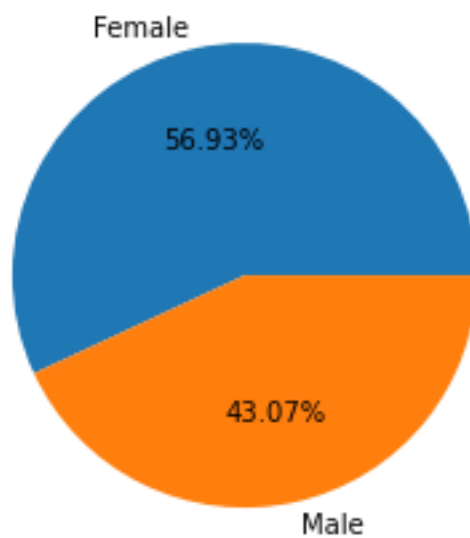
Bar Chart showing Number of People Suffer From heart disease:



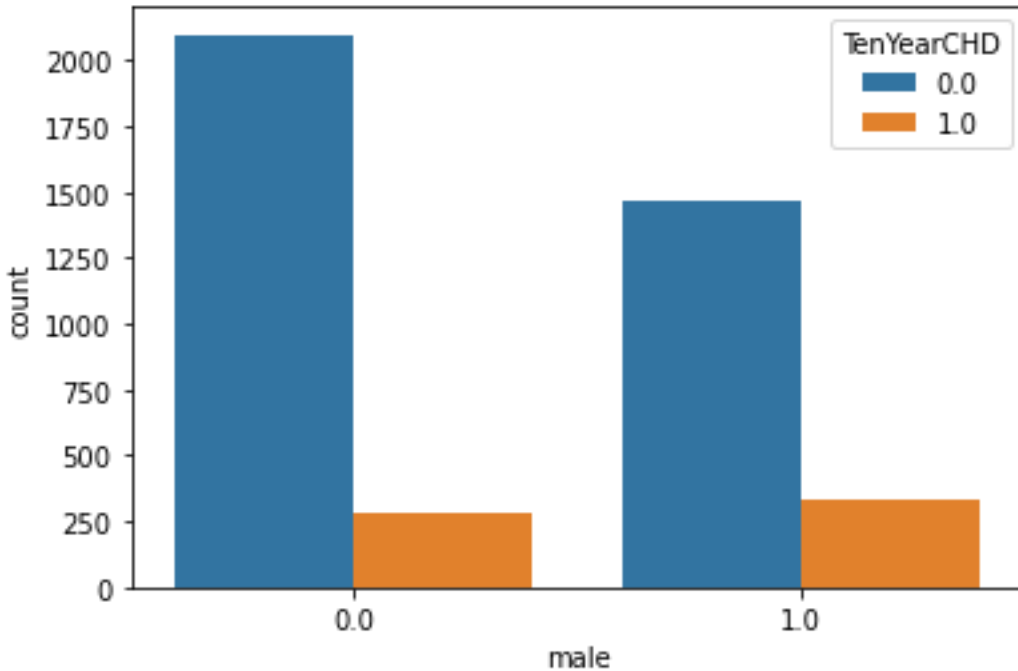
Pie Chart showing percentage of people with and without heart disease



Pie chart showing male and female percentage



Male and Female having disease or not



5.2.3 Splitting dataset

Data set is splitted into training and testing subsets .

For training 80% data and for testing 20% of data is selected .

5.2.4 Algorithms

For building machine learning model following algorithms

Are used.

A. ANN

B.1D CNN

C.UNET

5.2.5 Performance Evaluation Criteria

Commonly, the performance of DL prediction systems is frequently assessed using metrics based on the classification algorithm. The accuracy is used to evaluate the prediction findings in this research.

6.GUI Implementation

A GUI is developed in python by using tkinter package to generate a simple interface which takes input for all the values necessary for evaluation. After the input is taken from the user, prompt appears which decides whether a person has a presence of heart disease or not.

CORONARY HEART DISEASE PREDICTION USING DEEP LEARNING

ANN MODEL CNN 1D MODEL UNET MODEL MODEL COMPARASION

MALE [Input Field] AGE [Input Field]

CIGARETTE PER DAY [Input Field] BP MEDICINE [Input Field]

HEART STROKE IN PAST [Input Field] HYPERTENSION [Input Field]

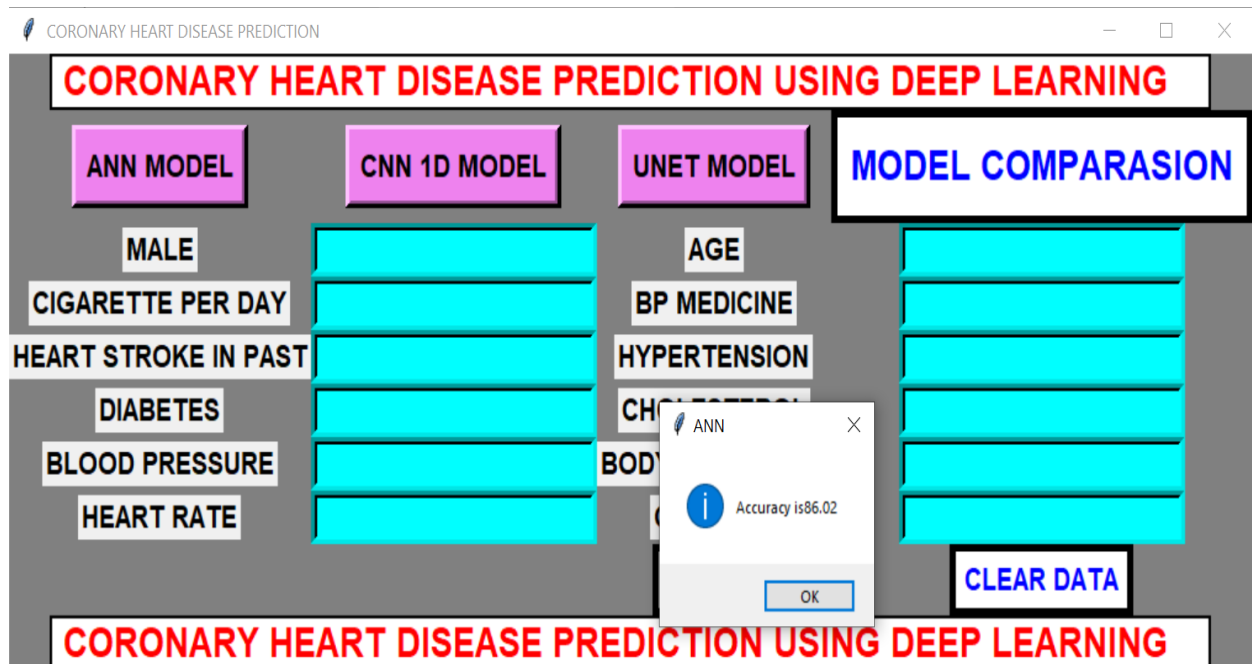
DIABETES [Input Field] CHOLESTEROL [Input Field]

BLOOD PRESSURE [Input Field] BODY MASS INDEX [Input Field]

HEART RATE [Input Field] GLUCOSE [Input Field]

SUBMIT CLEAR DATA

CORONARY HEART DISEASE PREDICTION USING DEEP LEARNING



Accuracy of ANN Model

CORONARY HEART DISEASE PREDICTION USING DEEP LEARNING**ANN MODEL****CNN 1D MODEL****UNET MODEL****MODEL COMPARASION****MALE**

1

CIGARETTE PER DAY

0

HEART STROKE IN PAST

0

DIABETES

1

BLOOD PRESSURE

100

HEART RATE

90

AGE

45

BP MEDICINE

0

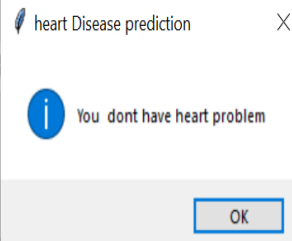
HYPERTENSION

0

00

8

00

**CLEAR DATA****CORONARY HEART DISEASE PREDICTION USING DEEP LEARNING**

CORONARY HEART DISEASE PREDICTION USING DEEP LEARNING

ANN MODEL	CNN 1D MODEL	UNET MODEL	MODEL COMPARASION
MALE	1	AGE	59
CIGARETTE PER DAY	20	BP MEDICINE	1
HEART STROKE IN PAST	1	HYPERTENSION	1
DIABETES	1	CHOLESTEROL	200
BLOOD PRESSURE	100	GLUCOSE	0
HEART RATE	120		
			CLEAR DATA

CORONARY HEART DISEASE PREDICTION USING DEEP LEARNING

heart Disease prediction×

i

You may suffer from heart problem

7. conclusion

- In this project 3 different deep learning algorithms are applied for prediction of heart disease.
- They are ANN ,1d CNN,UNET algorithm,
- From the above algorithms ANN has highest accuracy. So the best model for coronary heart disease prediction is ANN.

8. Future Work

- In this project 3 machine learning algorithms are applied for prediction of coronary heart disease .It was found ANN performs better than other algorithms. Also a GUI is developed for prediction of heart disease. In future a web based system will be developed so that people from different parts of world can use it.

9.Reference

- [1] Heart Disease Prediction Using Machine Learning and Data Mining Techniques: Application of Framingham Dataset. Turkish Journal of Computer and Mathematics Education. Vol.12 No.14(2021), 4864- 4870. Walaa Adel Mahmoud , Prof. Dr. Mohamed Aborizka ,Prof. Dr. Fathy Ahmed Elsayed Amer2 .
- [2] J.S.Sonawane and D.R.Patil, "Prediction of heart disease using multilayer perceptron neural network," International Conference on Information Communication and Embedded Systems (ICICES 2014), pp.1–6, Feb, 2014. doi:10.1109/ICICES.2014.7033860.
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